

Resolving the Quantum Measurement Gap

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1. Introduction: The Quantum Measurement Gap

In standard quantum mechanics, the evolution of a system is governed by the Schrödinger equation, which is linear, deterministic, and unitary:

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$$

However, during a measurement, the system appears to abruptly "collapse" into an eigenstate of the observable, violating this smooth evolution. This discontinuity between deterministic evolution and probabilistic measurement is the **quantum gap**.

Traditionally, interpretations like Copenhagen (collapse), Many-Worlds (branching), and decoherence attempt to address this, but none resolve the core problem within a fully physical, unified field-theoretic context.

2. The Resonance–Torsion Solution: Physical Overview

This framework eliminates the need for collapse by introducing a physically grounded dynamical mechanism:

Measurement is a process of resonance lock-in

When an observer interacts with a quantum system, the scalar field $\phi(x,t)$ (which permeates spacetime and couples to matter) dynamically responds to ambient energy, spin, and information gradients. This scalar field is coupled to spacetime torsion $T_{\mu\nu\lambda}$, an antisymmetric geometric field capturing intrinsic spin density and rotational degrees of freedom in the affine connection.

The measurement event triggers a **resonance cascade** between ϕ , $T_{\mu\nu\lambda}$, and the observer information gradient $\nabla_\mu I(x)$, causing the system to converge to a stable eigenmode rather than stochastically collapsing.

3. The Mathematical Structure

3.1 Unified Action Functional

$$S = \int d^4x \sqrt{-g} \mathcal{L}_{\text{total}}$$

Where the Lagrangian includes:

$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{gravity}} + \mathcal{L}_\phi + \mathcal{L}_{\text{torsion}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{feedback}} + \dots$

$$\mathcal{L}_{\text{gravity}} = \frac{1}{16\pi G} (R + \alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \gamma R_{\square})$$

3.2 Key Terms

Gravity:

$$\mathcal{L}_{\text{gravity}} = \frac{1}{16\pi G} (R + \alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \gamma R_{\square})$$

Scalar Field (ϕ):

$$\mathcal{L}_\phi = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi)$$

$$V(\phi) = \frac{1}{2} m^2 \phi^2 + \lambda \phi^4 + \xi \phi T$$

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Torsion:

$$\mathcal{L}_{\text{torsion}} = \eta S_{\mu\nu\lambda} S^{\lambda\mu\nu}, \quad T_{\mu\nu\lambda} = S_{\mu\nu\lambda}$$

Observer Feedback:

$$L_{\text{feedback}} = \zeta \phi T_{\lambda\mu\nu} T^{\lambda\mu\nu} + \chi g^{\mu\nu} \nabla_\mu I \nabla_\nu I + \tau g^{5\mu} J_\mu u_5$$
$$L_{\text{feedback}} = \zeta \phi T_{\lambda\mu\nu} T^{\lambda\mu\nu} + \chi g^{\mu\nu} \nabla_\mu I \nabla_\nu I + \tau g^{5\mu} J_\mu u_5$$

4. The Measurement Mechanism

4.1 Field Equation for $\phi\phi$

$$\square \phi - m^2 \phi = \lambda T + \xi T_{\mu\nu\lambda} T^{\mu\nu\lambda} + \dots$$
$$\square \phi - m^2 \phi = \lambda T + \xi T_{\mu\nu\lambda} T^{\mu\nu\lambda} + \dots$$

The $\phi\phi$ -field responds to mass-energy (TT), spin (via torsion), and resonance terms. It behaves as a field that converges to eigenmodes under environmental influence.

4.2 Resonance Lock-in

Measurement corresponds to $\phi\phi$ converging to a mode $\phi_n\phi_n$ that satisfies the coupled system:

$$\square \phi_n = \lambda T + \text{torsion feedback} \Rightarrow \phi \sim \phi_n$$
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where the eigenmode is selected by boundary and spectral conditions.

This is mathematically supported by a Fredholm determinant structure:

$$\det(I - H(s)) = \prod_{\delta} \zeta(s + \delta)^2$$
$$\det(I - H(s)) = \delta \prod \zeta(s + \delta)^2$$

indicating that all physical fields are constrained by number-theoretic resonance spectra.

5. Resolution of the Measurement Problem

In summary:

- The observer interacts with the system via information gradients $\nabla_\mu I(x) \nabla_\mu I(x)$.
- The scalar $\phi\phi$ and torsion $T_{\mu\nu\lambda} T^{\mu\nu\lambda}$ fields respond dynamically.
- This modifies spacetime via $g_{\mu\nu} \sim \nabla I \nabla I g_{\mu\nu} \sim \nabla I \nabla I$.
- Rather than collapse, the system locks into a resonant eigenstate.
- This process respects energy conservation and unitarity.

There is no stochastic projection — measurement is a deterministic resonance convergence governed by physical fields.

6. Conclusion

This framework:

- **Restores determinism and field causality** to quantum mechanics.
- **Integrates GR, QFT, and quantum measurement** into one action.
- **Proposes observable fifth-force effects**, vacuum polarization, and quantized coupling corrections.

The quantum gap is closed not by interpretation, but by explicit field dynamics, anchored in geometry, torsion, scalar flows, and observer interaction.